What Graph-Geodesic Embeddings Preserve: Path Fidelity, Scale, and Geometric Recovery

Pawel Gajer and Jacques Ravel

Author review draft, built 2026-09-06 19:37:10 EDT
Source commit: `48e76a6`

Abstract

Embedding a neighborhood graph in Euclidean space can reproduce its distances accurately while misrepresenting the geometry the graph was built to approximate. Multidimensional scaling (MDS) fits graph distances with chords between embedded points, whereas path criteria compare the lengths of retained graph routes before and after embedding. What either kind of agreement establishes about recovery of the sampled coordinates requires a separate analysis. We examine these relationships through theory and controlled comparisons of classical scaling, metric stress MDS, and edge-length refinement on paraboloids and saddles, evaluating retained paths with the graph-geodesic MDS (GMDS) criterion. Uniform geodesic bounds show that expanding paraboloids approach a line metric after normalization and saddles a four-branch tree metric; under a sample-rank condition, the saddle's classical-MDS limit spans three Euclidean dimensions although the tree is intrinsically one-dimensional. Edge refinement lowers retained-path error in all 25 bounded-saddle settings under both MDS initializers, but not in every setting of a larger radius study, and does not consistently lower coordinate error. Profiling a common target scale in an unnormalized edge loss lets every positive-loss configuration reduce the objective by contraction alone; fixed-scale controls separate this effect from changes in shape. Path agreement, endpoint agreement, and coordinate recovery therefore require separate assessment under an explicit scale convention; agreement on one does not in general certify the others.

1 Introduction

Data-derived graphs represent relationships in settings ranging from scientific collaboration networks (Newman, 2001) to structural brain connectivity (Hagmann et al., 2008) and single-cell expression data (Wolf et al., 2019). Their vertices and edges describe different scientific objects, but drawing any of them in Euclidean space raises a common question: which relationships should the coordinates preserve? Observed connections, similarity weights and geometric lengths have different interpretations, so a graph's scientific use determines which measurements its drawing must support.

For observations sampled from a curved structure, a neighborhood graph approximates intrinsic distances: nearby observations are joined by edges, and shortest paths accumulate edge lengths to estimate distances along the structure. Isomap constructs such a graph from nearest-neighbor or radius neighborhoods, computes its shortest-path distances, and applies classical multidimensional scaling (MDS) to obtain Euclidean coordinates (Tenenbaum et al., 2000). The construction involves two approximations: the graph approximates the intrinsic geometry, and the coordinates approximate the graph. An accurate embedding of the graph therefore cannot establish that the graph captured the geometry of interest.

The embedding step also changes what is measured. MDS seeks coordinates whose endpoint separations approximate the supplied dissimilarities (Mair et al., 2022; R Core Team, 2026); when

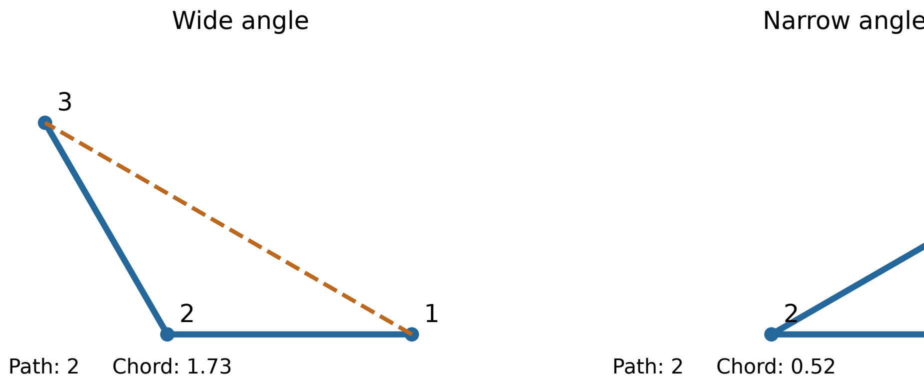

Figure 1: Two drawings of the same three-vertex unit path. Blue segments retain length one, and the route from 1 through 2 to 3 retains length two. The orange endpoint chord changes with the angle. Both drawings preserve every edge length and every retained path length. All spatial units are equal.

those dissimilarities are graph shortest-path distances, intrinsic path lengths become targets for Euclidean chords. This choice of fidelity criterion coincides with intrinsic distance preservation only when the target geometry has an exact Euclidean realization, as in the intrinsically Euclidean domains covered by Isomap's recovery result (Tenenbaum et al., 2000). On more general curved structures the two criteria separate, as they do for a Riemannian isometric embedding, which preserves the lengths of curves on the manifold while Euclidean chords between its points can be shorter (Gorodski, 2022, Sections 1.2–1.3 and 3.2). The curve-length viewpoint suggests a graph criterion of the same kind: retain a selected route between two vertices and compare its length before and after drawing, so that both measurements refer to the same sequence of edges.

Graph-geodesic multidimensional scaling (GMDS), as considered here, compares these retained-path lengths with their input lengths. It is a discrete, finite-path analogue of Riemannian curve-length preservation, and the analogy has limits: the graph supplies no full tangent-space metric, and path-length constraints need not prevent overlapping configurations, so a small path loss is not a certificate of isometry or geometric recovery. Because exact preservation of every edge length preserves every path length on the same graph, edge-length refinement of Kamada–Kawai type (Kamada and Kawai, 1989), referred to here as edge-KK, is one route to path agreement, although on a flexible graph exact path agreement still leaves the configuration free to move (Figure 1). Whether edge lengths also determine the configuration up to rigid motion is the global-rigidity problem (Gortler et al., 2010).

Paraboloids and saddles allow these questions to be studied against a known reference geometry. As the sampled domain expands, we derive uniform bounds showing that normalized paraboloid geodesics approach a line metric and saddle geodesics approach a four-branch tree metric. The tree is intrinsically one-dimensional, yet under an explicit sample-rank condition its classical-MDS approximation spans three dimensions, so the dimension of an intrinsic distance structure need not equal the Euclidean span of its embedding. These results concern particular surface limits within the broader study of classical MDS on metric spaces (Lim and Mémoli, 2024).

Two controlled studies connect these limits to achieved embeddings. The first compares classical and stress-MDS initializations followed by edge-KK on a bounded saddle; the second varies domain radius, neighborhood size, graph construction and target scale on both surfaces. Together they separate changes in endpoint and retained-path agreement from changes in coordinate recovery.

Elementary edge-to-path bounds and least-squares scale identities clarify what the objectives measure; in particular, profiling the target scale in an unnormalized edge loss precludes positive-loss local minima whenever uniform contraction is admissible, and fixed-scale controls separate this degeneracy from improvements in relative geometry. The principal theoretical contributions are the surface-specific uniform bounds and the four-branch Gram factorization; the experiments document the effects of graph, initialization and scale choices without assuming a universal ordering of embedding methods. The analytical statements have fixed-sample assumptions, and the numerical comparisons concern the stated samples and optimization budgets.

2 Distances, objectives, and scale

2.1 A fixed graph and a fixed family of routes

To separate the effect of embedding from the effect of graph construction, hold the input graph fixed: a connected undirected graph $G = (V, E, \ell)$ with n vertices, indexed by $1, \ldots, n$, and positive edge lengths ℓ_e. Its shortest-path distance between vertices i and j is $d_G(i, j)$. For each unordered pair, choose one shortest path γ_{ij}, breaking ties deterministically, and retain these routes as the family Γ.

An embedding assigns each vertex a point $z_i \in \mathbb{R}^p$, collected in the coordinate matrix $Z = (z_1, \ldots, z_n)^T \in \mathbb{R}^{n\times p}$. The chord measures separation between endpoints; the retained-path length sums the embedded edges along the chosen route:

$$d_Z(i, j) = \|z_i - z_j\|, \qquad L_\Gamma(i, j; Z) = \sum_{(u,v)\in\gamma_{ij}} \|z_u - z_v\|. \tag{1}$$

The triangle inequality gives $d_Z(i, j) \leq L_\Gamma(i, j; Z)$. Equality requires the nonzero displacements along that path to have a common direction. This inequality compares measurements in the same configuration; it does not order their squared errors relative to an external target.

The route family is retained when the coordinates move. Recomputing shortest paths after replacing edge lengths by $\|z_u - z_v\|$ gives another quantity, which is no larger than the retained-route length. Because each pair keeps its original route, the retained-route lengths need not satisfy the triangle inequality after the vertices move.

The same edge can occur in many retained routes, so its length can affect several path errors. To express this dependence, collect the $|E|$ embedded edge lengths in $d_E(Z)$ and record route membership in the $\binom{n}{2} \times |E|$ incidence matrix A_Γ. Its (ij, e) entry is one if route γ_{ij} uses edge e, and zero otherwise. Collecting path lengths over unordered pairs gives $L_\Gamma(Z) = A_\Gamma d_E(Z)$ and $d_G = A_\Gamma \ell$. The sum of squared path-length errors, or raw fixed-path stress, is therefore

$$S_\Gamma(Z) = \|A_\Gamma(d_E(Z) - \ell)\|^2. \tag{2}$$

The shared edges couple the path residuals. Weighted edge stress generally has a different quadratic form in the edge residual vector.

The smooth counterpart clarifies what this path criterion measures. A Riemannian isometric embedding $\phi : (M, g) \to \mathbb{R}^p$ pulls the Euclidean metric back to the given metric, $\phi^* g_{\text{Euclidean}} = g$. For every piecewise smooth curve σ on M, its image has the same length:

$$L_{\text{Euclidean}}(\phi \circ \sigma) = L_g(\sigma).$$

Taking the infimum over curves gives intrinsic distances d_M on the domain and $d_{\phi(M)}$ on the image. Thus

$$\|\phi(x) - \phi(y)\| \leq d_{\phi(M)}(\phi(x), \phi(y)) = d_M(x, y), \qquad x, y \in M.$$

The inequality allows ambient shortcuts that are unavailable within the embedded manifold (Gorodski, 2022, Sections 1.2–1.3 and 3.2). For example, the standard unit circle is Riemannian-isometrically embedded in the plane, yet antipodal points have intrinsic distance π and chord distance 2.

In (2), the image of each retained graph route is the polygonal path through its embedded vertices. The criterion checks the total lengths of this finite route family, not every curve on an underlying manifold. Preserving all smooth curve lengths determines the Riemannian metric and hence angles; a graph route family does not provide that full information, and even exact agreement on selected routes need not preserve every edge length or make the drawing injective. Conversely, preserving every edge length preserves every route on the same abstract graph, with crossings in the drawing treated as crossings rather than additional junctions. This is the length-preservation connection underlying the GMDS criterion.

2.2 MDS and edge refinement

When the intended measurement is endpoint separation, each pair contributes its own distance error. For a symmetric $n \times n$ target dissimilarity matrix Δ, raw metric stress adds the squared errors of these chords:

$$S_\Delta(Z) = \sum_{i<j} (d_Z(i,j) - \Delta_{ij})^2. \tag{3}$$

Equal absolute distance errors make equal contributions to this unweighted loss, regardless of target distance. Equal relative errors make contributions proportional to Δ_{ij}^2. Long distances therefore matter more for a fixed relative discrepancy. This weighting alone does not decide whether a fitted surface becomes planar or linear. Section 3 derives different normalized input-metric limits for the paraboloid and saddle. For the saddle, three-dimensional span is proved for classical MDS under a sample-rank condition; the raw-stress fits provide finite-budget numerical evidence rather than the same guarantee.

Classical MDS approximates the geometry through inner products. Squaring the target distances entrywise and centering on both sides gives the matrix $B = -\frac{1}{2} J \Delta^{\circ 2} J$, where $J = I - \mathbf{1}\mathbf{1}^T/n$ and $\mathbf{1}$ is the n-vector of ones. Classical scaling retains up to the p largest positive eigenvalues and their eigenvectors. Its rank-constrained Gram approximation is commonly called strain (Lim and Mémoli, 2024, Theorem 2); classical scaling does not minimize (3).

After either MDS initialization, edge-KK refines the embedded edge lengths $d_e(Z)$. At a given stage, nonnegative stiffnesses κ_e weight their squared deviations from input lengths multiplied by a common target scale a:

$$E(Z,a) = \frac{1}{2} \sum_{e \in E} \kappa_e (d_e(Z) - a\ell_e)^2, \qquad \kappa_e \geq 0. \tag{4}$$

The experiments vary the stiffness schedule and compare a fixed target scale $a = 1$ with a scale profiled at each configuration. Both are edge objectives; neither directly minimizes (2). For a complete graph with direct-edge targets $\ell_{ij} = \Delta_{ij}$, uniform stiffness and fixed scale, $2E(Z,1) = S_\Delta(Z)$. This identity concerns edge lengths and endpoint chords. A retained route has the same embedded length as its endpoint chord when it is the direct edge; an alternative multiedge route tied in the input need not preserve that equality after fitting.

The use of an edge objective for path refinement rests on a simple bound: if every edge has small relative error at a common scale, every path does too.

Proposition 1 (Uniform edge-to-path control). *Suppose $a > 0$ and $|d_e(Z)/(a\ell_e) - 1| \le \epsilon$ for every edge. For any nonempty path γ in the same graph,*

$$\frac{|\sum_{e\in\gamma} d_e(Z) - a\sum_{e\in\gamma}\ell_e|}{a\sum_{e\in\gamma}\ell_e} \le \epsilon.$$

Proof. Bound the absolute sum of edge residuals by the sum of their absolute values, then use $|d_e - a\ell_e| \le \epsilon a\ell_e$ on each edge. □

The premise is a maximum relative error, not an average squared residual. Even zero edge error leaves geometric freedom on flexible graphs. Two unit edges sharing a vertex preserve all fixed-path lengths while their angle changes (Figure 1). At a zero angle their outer endpoints coincide. Exact path fidelity therefore need not imply an injective drawing. This counterexample concerns a flexible graph; it does not establish continuous flexibility of the denser graphs in the experiments.

2.3 Profiled scale and contraction

Fitting a common target scale allows the input lengths to match the overall size of the drawing. The effect on optimization becomes visible by expanding the edge loss. The weighted squared sizes of the embedded and input edges are $S_{dd} = \sum_e \kappa_e d_e^2$ and $S_{\ell\ell} = \sum_e \kappa_e \ell_e^2$; their weighted cross-product is $S_{d\ell} = \sum_e \kappa_e d_e \ell_e$. If $S_{\ell\ell} > 0$, the best target scale is $a(Z) = S_{d\ell}/S_{\ell\ell}$. Substitution into (4) gives the profiled loss

$$\mathcal{E}(Z) = \min_a E(Z,a) = \tfrac{1}{2}\big(S_{dd} - S_{d\ell}^2/S_{\ell\ell}\big). \tag{5}$$

Proposition 2 (Homogeneity of the profiled loss). *Hold the stiffnesses fixed and assume $S_{\ell\ell} > 0$. For $c > 0$, $a(cZ) = ca(Z)$ and $\mathcal{E}(cZ) = c^2\mathcal{E}(Z)$. If $S_{dd} > 0$, the normalized quantity $Q = 2\mathcal{E}/S_{dd} = 1 - S_{d\ell}^2/(S_{dd}S_{\ell\ell})$ is invariant under uniform coordinate scaling.*

Proof. Scaling coordinates by c multiplies S_{dd} by c^2 and $S_{d\ell}$ by c, leaving $S_{\ell\ell}$ unchanged. Substitute these relations into (5). □

A configuration with positive profiled loss has a strict descent direction under uniform contraction. Thus no positive-loss local minimum exists in an unconstrained configuration domain admitting such contractions. Wherever the objective is differentiable, differentiating $\mathcal{E}(cZ) = c^2\mathcal{E}(Z)$ at $c = 1$ gives $\langle Z, \nabla\mathcal{E}(Z)\rangle = 2\mathcal{E}(Z)$, with the Frobenius inner product. A positive-loss stationary point is therefore impossible. If zero target scale and collapsed coordinates are admitted, total collapse attains zero loss. If scale must remain strictly positive, shrinking any configuration with $S_{d\ell} > 0$ still drives the infimum to zero, even when no noncollapsed zero-residual shape exists. These are properties of the unnormalized optimization problem at fixed stiffnesses; they do not assert that every finite run collapses. A small loss or gradient can reflect contraction without a change in relative shape. Separately normalized diagnostics remain meaningful.

For comparison, optimize the coordinate scale of a fixed shape against fixed edge targets. If $S_{dd} > 0$ and $S_{d\ell} > 0$, $\min_{c>0} E(cZ, 1) = S_{\ell\ell}Q/2$, attained at $c = S_{d\ell}/S_{dd}$. This is the weighted least-squares calculation underlying scale-normalized graph stress (Ahmed et al., 2026, Section 3.2), and it shows that a properly normalized shape loss and fixed-target fitting with optimal overall size have related shape objectives. The raw profiled loss (5) lacks this normalization; the additional issue here is profiling the target scale while minimizing the unnormalized objective over coordinates. The present experiments use fixed-target controls without introducing a new optimizer for Q.

3 Expanding surfaces and their metric limits

An elongated drawing can arise from the input distance geometry as well as from the fitting criterion. The paraboloid and saddle separate these effects because their smooth geodesic distances have explicit limits as the domain radius grows. On both surfaces, horizontal dimensions grow as r and heights as r^2. Dividing intrinsic distances by r^2 reveals which separations persist at the height scale.

3.1 Paraboloid: a normalized line limit

On the paraboloid $z = x^2 + y^2$, height depends only on distance from the vertical axis. To follow the same base locations as the domain expands, fix points $u_i \in \mathbb{R}^2$ in the unit disk. Their squared radii $q_i = \|u_i\|^2$ determine the surface points $P_i(r) = (ru_i, r^2 q_i)$. The intrinsic geodesic distance between two such points, restricted to the surface above the radius-r disk, is $g_{ij}(r)$. At the height scale, the distance is determined by vertical separation:

Proposition 3. *For every pair and $r > 0$,*

$$r^2|q_i - q_j| \leq g_{ij}(r) \leq r^2|q_i - q_j| + (1 + \pi)r.$$

Proof. Every surface path has length at least its absolute vertical displacement. For an upper bound, move radially from the outer endpoint to the inner radius, then along the shorter circular arc. Since $\sqrt{1 + 4\rho^2} \leq 1 + 2\rho$, the radial part is at most the height difference plus r, and the arc is at most πr. This path stays in the disk. □

The normalized distances converge uniformly to $|q_i - q_j|$. Angular separation vanishes at this scale; this does not mean its unscaled distance vanishes.

Theorem 4. *Fix a finite sample with at least two distinct q_i and a dimension $p \geq 1$. Every global raw-stress minimizer for $g(r)$, after division of coordinates by r^2, has a centered Gram matrix converging to $(Jq)(Jq)^T$. The same holds for classical MDS and for approximate raw-stress minimizers with excess stress $o(r^4)$.*

Proof. Writing $Z = r^2 V$ divides the stress by r^4 and the targets by r^2. The candidate $V_i = (q_i, 0, \ldots, 0)$ has normalized stress at most $\binom{n}{2}(1 + \pi)^2/r^2$. Minimizers, including the stated approximate ones, therefore have residuals tending to zero. Their pair distances converge to $|q_i - q_j|$, and double-centering their squared distances gives the Gram limit. For classical MDS, $B(r)/r^4$ converges directly to the same rank-one matrix; its unique positive eigenvalue is separated from zero. □

This is a fixed-sample result. It does not establish that the fitted optimizer outputs attain its approximation requirement. For coupled samples whose $q_i(r)$ converge, the same proof uses the moving line candidate and requires at least two distinct limiting heights.

3.2 Saddle: a tree metric with three-dimensional classical span

On the saddle, equal height does not always allow a connection that stays at that height. Positive levels have two components, as do negative levels. This separation determines the limiting distances. Fix $u_i = (x_i, y_i)$ in the unit disk and write the corresponding points on $z = x^2 - y^2$ as $P_i(r) = (rx_i, ry_i, r^2 h_i)$, where $h_i = x_i^2 - y_i^2$ and $t_i = |h_i|$. Positive heights have separate $+x$ and $-x$ branches; negative heights have separate $+y$ and $-y$ branches. Zero heights map to a common root. Define the four-branch tree distance

$$T_{ij} = \begin{cases} |t_i - t_j|, & i, j \text{ on the same branch,} \\ t_i + t_j, & i, j \text{ on different branches.} \end{cases} \tag{6}$$

Proposition 5. *The intrinsic saddle distances on the disk-restricted patch satisfy* $r^2 T_{ij} \le g_{ij}(r) \le r^2 T_{ij} + 4r$.

Proof. Length dominates total vertical variation. Different signs of height require crossing zero. Opposite positive branches require x to cross zero, where height is nonpositive; opposite negative branches similarly require y to cross zero. These facts give the lower bound.

For the upper bound, follow each constant-height hyperbola to its corresponding coordinate semiaxis. On a positive branch, $|dx/dy| \le 1$ and $|y| \le r/\sqrt{2}$, so each connector has length at most r; negative branches exchange x and y. Connect the semiaxis points monotonically on one branch or through the origin for different branches. The inequality used in Proposition 3 bounds this segment by the required vertical variation plus $2r$. All segments remain inside the disk. Zero-height points follow a straight ruling to the origin. □

Thus $g(r)/r^2 \to T$ uniformly. The limit is intrinsically a one-dimensional tree. Its classical Euclidean approximation need not be one-dimensional.

To determine the Euclidean span selected by classical MDS, encode each point by its distance from the root in its branch coordinate. In the next statement, e_b denotes the bth standard basis vector of $\mathbb{R}^4$.

Theorem 6. *Let F be the $n \times 4$ matrix with row $t_i e_{\beta(i)}^T$ for a nonroot point, where $\beta(i)$ is its branch label, and zero row for a root point. If JF has rank four, the double-centered Gram matrix $-\frac{1}{2} J T^{\circ 2} J$ has exactly three positive eigenvalues and one negative eigenvalue. Classical three-dimensional MDS of the expanding saddle therefore retains three nonzero limiting singular values after coordinate division by r^2.*

Proof. Since $t = F\mathbf{1}$, expansion of (6) and double-centering give

$$B_T = -\tfrac{1}{2} J T^{\circ 2} J = JF(2I_4 - \mathbf{1}\mathbf{1}^T)F^T J. \tag{7}$$

The middle matrix has eigenvalues $2, 2, 2, -2$. The full column rank of JF transfers these nonzero inertia counts to B_T. Proposition 5 implies $B(r)/r^4 \to B_T$, so the three positive classical directions persist. □

A sufficient rank condition is one nonroot observation per branch and two distinct positive lengths on at least one branch. A root plus one nonroot point per branch also suffices. Missing branches or degenerate samples need separate analysis. The theorem applies to classical scaling; it does not exclude every planar minimizer of raw stress on the tree metric. Non-Euclidean Gram spectra for tree metrics are already illustrated by the three-leaf-star example of Lim and Mémoli (2024, Example 2.6); the additions here are the surface distance bound that identifies the tree arising from the expanding saddle and the rank condition under which its classical approximation retains three positive directions.

A simple example makes the spatial dimension concrete. Four branch tips at unit distance from the root have pairwise distances two. If only these tips are observed, their centered matrix is $2J$, and classical MDS represents them exactly as a regular tetrahedron. A planar cross has the right tree routes but not these Euclidean endpoint distances. This tip-only sample has rank three in JF and illustrates why the rank-four condition above is sufficient, not necessary, for a three-dimensional classical span.

Gaussian curvature alone does not resolve this comparison. On these two surfaces, curvature is respectively $4/(1+4\rho^2)^2$ and $-4/(1+4\rho^2)^2$. Their magnitudes decay identically with radius, yet

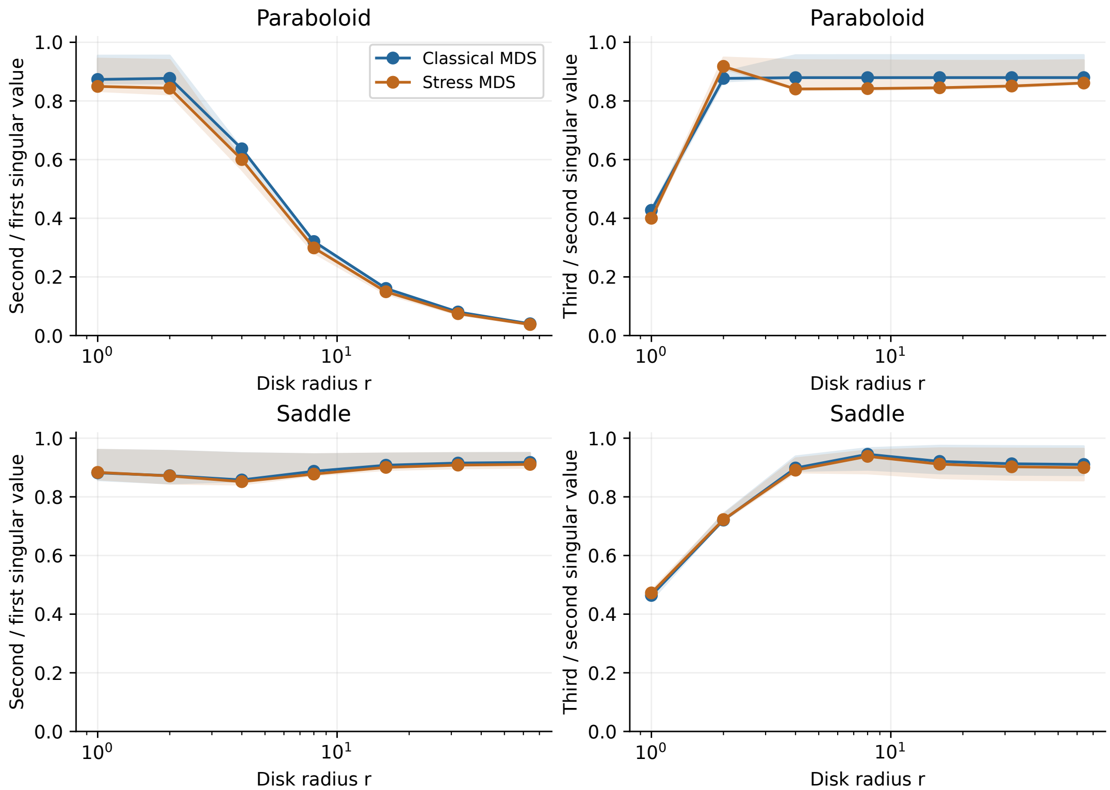

Figure 2: The expanding paraboloid becomes elongated, while the saddle retains three comparable embedding directions. Shape ratios are shown for complete geodesic graphs ($n = 240$, $k = 239$), under base-disk sampling. Curves show medians and bands the observed ranges across three samples. Both ratios are needed: two small transverse directions can remain comparable while the configuration approaches a line. These are classical and finite-budget stress-MDS fits to strict graph distances; numerical smooth-geodesic targets and their strict distances can differ slightly. The classical and global raw-stress statements have the separate assumptions in Theorems 4 and 6.

their normalized distance limits differ because of the connectivity of height levels. Figure 2 shows the corresponding finite-sample behavior.

This explanation can be stated using Reeb graphs. On the unit disk, collapse each connected level component of $q(u) = \|u\|^2$ or $h(u) = x^2 - y^2$, $u = (x, y)$, to a point. Give a path in the quotient its total variation in function value, as in the metric Reeb construction (Mémoli et al., 2023, Definitions 2.1 and 3.1). Both height functions are 1-Lipschitz for their normalized surface distances. For q, the quotient is $[0, 1]$ with ordinary distance. For h, positive and negative levels each have two components, joined at the connected zero level; the quotient is the four-branch tree in (6). Propositions 3 and 5 supply explicit uniform distance errors for these particular quotients. The identification uses this established construction, without requiring a general collapse theorem.

4 Controlled experiments

The surface limits describe intrinsic distances and idealized MDS behavior; the two studies described next examine what happens when a sampled neighborhood graph and a finite optimization budget intervene. Classical scaling uses `cmdscale` in R (R Core Team, 2026); stress MDS uses the ratio-

Table 1: The two controlled studies differ in design and are not pooled. Details and controls are given in Sections 4.1 and 4.2.

	Bounded-saddle initializer comparison	Radius, graph and scale comparison
Surface, domain	$z = 0.8(x^2 - y^2)$ over $[-1, 1]^2$	$z = x^2 \pm y^2$ above disks of radius $r \in \{1, 2, 4, \ldots, 64\}$
Samples	five, $n = 1{,}000$, area-uniform	per surface/sampling combination: three $n = 240$ replicates and one nested $n = 480$ replicate; base-disk and surface-area uniform
Graphs	symmetric-union kNN, ambient chords; 25 graphs	symmetric-union kNN, geodesic or ambient targets; $k \in \{4, 8, \ldots, 128, 239\}$
Initializers	classical; stress (six starts)	classical; stress (three starts); fixed full-geodesic classical control
Edge-KK scale	profiled	profiled and fixed
Diagnostics reported	retained-path, edge, chord and coordinate errors	path-to-surface and coordinate errors, shape ratios, calibration factor; component errors for selected fits

SMACOF implementation in `smacof` (Mair et al., 2022), with the package adapter restoring physical target units; graph construction, edge-KK and the diagnostics use the `grip` R package (Data and code availability). Table 1 summarizes the two designs. They differ in domain, sample size and scale policy, so each comparison retains its own samples, targets and scale policy.

4.1 Bounded-saddle initializer comparison

The first study uses five independent area-uniform samples of $n = 1{,}000$ points on $z = 0.8(x^2 - y^2)$ over $[-1, 1]^2$. Symmetric-union kNN graphs use ambient chord lengths both for neighbor selection and as edge targets. The selected neighborhood sizes are 71, 70, 67, 72 and 73, based on comparison with numerical surface-geodesic references. This calibration minimizes unscaled graph-to-reference relative RMSE over $k = 3, \ldots, 80$, using 119,744 unordered pairs obtained from 128 reference sources and all targets. It uses the known surface and is not a neighborhood-selection rule for unknown biological geometry. Each sample additionally uses $k = 32$, $k = 80$, and the selected $k \pm 5$, giving 25 graphs. These graphs are connected without added bridges.

Classical and raw-stress MDS use strict graph shortest-path matrices and three output dimensions. Stress MDS selects the least achieved stress from six starts: one classical and five Gaussian, with 1,000 iterations per start. Both branches then receive the same five-stage edge-KK schedule, interpolating from density stiffness to uniform stiffness with 200 steps per stage. This study predates the fixed-scale controls of the radius study and uses the profiled target scale throughout. The Gaussian draws are paired across k within each sample. Additional iterations are controls and do not replace the primary fits.

4.2 Radius, graph, and scale comparison

The second study samples $z = x^2 + y^2$ and $z = x^2 - y^2$ above disks with $r \in \{1, 2, 4, 8, 16, 32, 64\}$. For each surface, both uniform base-disk and uniform surface-area sampling are considered, with three independent $n = 240$ samples and coupled quantiles/angles across radii. No uniform probability distribution on the entire infinite-area surface is assumed.

The neighborhood grid is $k \in \{4, 8, 16, 32, 64, 128, 239\}$. Separate graph regimes use either numerical smooth geodesics or ambient chords for both ranking and edge targets. Graphs are symmetrized by union; minimum-spanning-tree edges repair disconnected cases and are recorded. Classical and stress MDS use each graph's strict distances. Stress MDS uses one classical and two Gaussian starts, each limited to 1,000 iterations. In the geodesic regime, a full-surface-distance classical initializer held fixed across k is an additional control.

Every initializer receives both density continuation and an equal-budget uniform-only refinement, under both profiled and fixed target scales. The primary schedule has the same five mixing values and 1,000 total steps as the first study. Numerical inputs are scaled by the RMS smooth distance within each case and outputs are restored to physical units. The fixed target scale is one in these normalized fitting units.

A nested $n = 480$ sample at $r = 64$ retains the original 240 points of replicate one and uses $k \in \{8, 16, 64, 128, 479\}$. Perturbation, additional-iteration, random-start and original-coordinate controls use 32 selected high-radius settings. Shared original/random controls appear in more than one branch comparison and do not supply independent replications.

4.3 Diagnostics and numerical validation

Separating relative geometry from overall size requires an explicit scale convention for each diagnostic. For a positive target vector t and an achieved length vector y, the least-squares calibration of the targets is $b = \langle y, t\rangle/\|t\|^2$. Edge and retained-path relative errors are $\|y - bt\|/\|bt\|$, each with its own diagnostic calibration. Chord Stress-1 uses $\|y - bt\|/\|y\|$. These scale conventions are explicit because normalizations called stress are not interchangeable (Mair et al., 2022). Graph-to-surface distance error uses common physical units. Embedded path-to-surface error uses the calibration estimated from graph edges, so it does not fit a new scale against the surface reference.

Recovery of the generating configuration requires a separate comparison of coordinates. Coordinate error is a normalized Procrustes residual after translation, orthogonal transformation and one uniform scaling, with vertex correspondence retained; it is a residual, not a classification rate. Singular values $s_1 \geq s_2 \geq s_3$ distinguish elongation (s_2/s_1 small) from planarity relative to the second direction (s_3/s_2 small). Neither is a certificate of topology or surface recovery. To assess horizontal and vertical recovery separately, we also split the residual from this same global similarity alignment into its (x, y) and z components, dividing each norm by the corresponding centered truth norm. No additional component-specific rotation or scale is fitted. These errors reveal distortions that the global norm can conceal when height dominates.

The radius study contains 1,216 primary graphs, 16,416 scored primary configurations and 3,648 MDS starts; 74 graphs require connectivity repair. All 88 numerical input matrices pass symmetry, chord lower bounds, constructed surface-path upper bounds and exhaustive vertex-triple checks within 10^{-7} times RMS distance. Independent high-radius numerical validation covers 256 actual pairs in 16 cases, with maximum sampled relative discrepancy 5.16×10^{-8} over those sampled pairs. Independent reconstruction from 16,928 primary and control coordinate arrays checks singular-value quantities, edge calibration and error, raw chord stress, chord Stress-1, coordinate error, and graph-to-surface error. These diagnostics agree to 2.59×10^{-12} after dimensional scaling; this full validator does not reconstruct retained-path scores. Path validation is separate: fitting-time checks compare with the package scorer and explicitly sum 31 routes per configuration. An additional ordinary-R calculation sums all 28,680 retained routes in each of 24 configurations at $n = 240$, $r = 64$, $k = 32$, spanning both surfaces, three base-disk replicates, both graph regimes, and stress

Table 2: Path agreement improves after edge-KK while chord agreement worsens: bounded-saddle medians over five selected graphs. Retained-path and edge columns use separately calibrated relative errors; chord is profiled Stress-1; coordinate error uses similarity alignment. Edge-KK uses the profiled target scale.

Method	Retained path (%)	Edge (%)	Chord (%)	Coordinate (%)
Classical MDS	3.451	7.133	3.459	29.19
Stress MDS	2.264	5.081	2.803	23.20
Classical MDS + edge-KK	0.245	0.382	4.715	30.67
Stress MDS + edge-KK	0.164	0.227	5.213	22.32

fits before and after fixed-scale density continuation. Retained-path and path-to-surface errors agree with the frozen scores to 7.60×10^{-15} in absolute error units. The retained near-tie routes and strict matrices are recorded separately.

5 Results

Counts and medians summarize the declared, partly coupled graph grids at finite optimization budgets. They describe these settings without implying independent replication, population uncertainty estimates or global optimality.

5.1 Path improvement and geometric agreement

In the bounded-saddle comparison, refinement improves path agreement while worsening normalized chord agreement. Edge-KK decreases retained-path error in all 25 bounded-saddle settings under both initializers and increases chord Stress-1 in all corresponding comparisons. At the selected graphs, median path error decreases from 3.451% to 0.245% for classical initialization and from 2.264% to 0.164% for stress initialization (Table 2, Figure 3). Stress initialization gives the lower final path error in 23 of 25 graphs; classical initialization gives the lower error on the remaining two graphs.

Coordinate agreement does not follow the same pattern. The median coordinate error increases under classical initialization, while decreasing slightly under stress initialization. Across selected graphs, stress-initialized refinement gives coordinate errors ranging from 4.60% to 27.54% despite small final path errors. All recorded edge-KK stages reach their iteration limits.

5.2 Graph fidelity and coordinate recovery separate on the saddle

The large-radius saddle separates intrinsic-distance agreement from coordinate recovery most clearly. At $r = 64$, $k = 32$, the fixed-scale stress-initialized geodesic saddle fits have median path-to-surface error 0.36% and coordinate error 80.38%, whereas ambient-graph fits give 0.90% coordinate error but 37.00% path-to-surface error (Table 3). The two graph regimes therefore answer different geometric questions. Figure 4 shows the corresponding configurations for the first sample, together with the smaller-radius comparison. The narrow equal-axis views at large r reflect the surface's ambient aspect ratio, which does not make the intrinsic saddle distance metric a line metric.

Height dominates the global coordinate norm on both surfaces at this radius. For the ambient saddle, the median horizontal error after the same alignment is 35.24%, despite a global error of 0.90% and a vertical error of 0.03% (Table 4). Simply setting the original horizontal coordinates to zero gives a height-only saddle configuration with median global residual 2.64%, even before

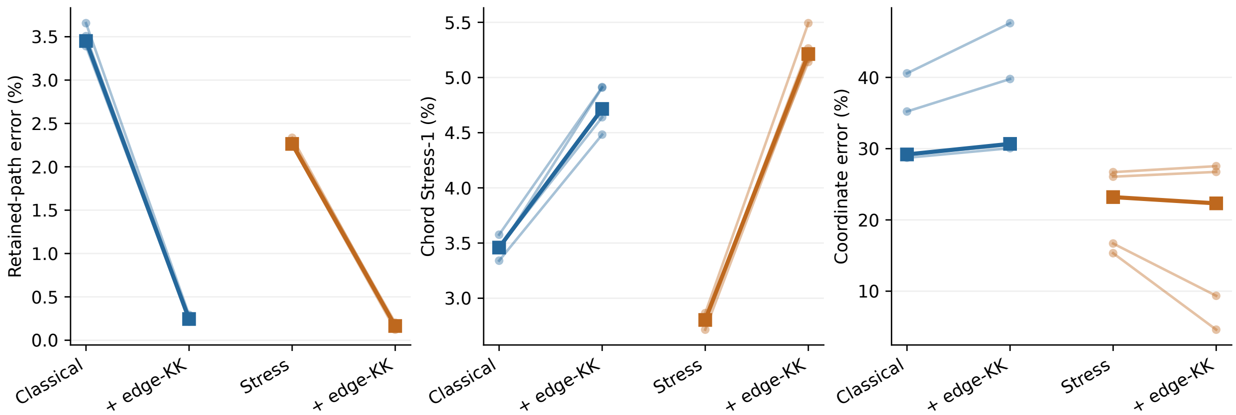

Figure 3: Path, chord and coordinate errors respond differently to edge-KK. Paired bounded-saddle comparisons use the selected graphs. Thin lines join the same sample before and after edge-KK; squares show medians over five samples. Blue denotes classical initialization and orange stress initialization. Vertical units and ranges differ across the three diagnostics.

Table 3: Graph construction separates path agreement from coordinate recovery on the saddle. Radius-study medians over three base-disk samples at $r = 64$, $k = 32$, using stress MDS followed by fixed-scale density continuation. Retained-path/surface error uses edge calibration; coordinate error uses similarity alignment.

Surface	Graph	s_2/s_1	s_3/s_2	Coordinate (%)	Retained path/ surface (%)
Paraboloid	geodesic	0.038	0.846	1.46	0.34
Paraboloid	ambient	0.028	0.915	0.88	0.38
Saddle	geodesic	0.901	0.896	80.38	0.36
Saddle	ambient	0.019	0.917	0.90	37.00

optimizing its alignment. The ambient fits improve on that baseline, but their small global residual does not certify recovery of the sampled disk. The paraboloid fits are strongly elongated under both graph regimes: their median horizontal errors are 23.66% for ambient graphs and 38.58% for geodesic graphs. Component errors, as well as the shape ratios, are needed to assess transverse geometry.

5.3 Neighborhood and initialization sensitivity

Changing the neighborhood size changes both the graph approximation and the constraints imposed during refinement. Graph-to-surface and embedded-path-to-surface errors vary with k (Figure 5). A small graph-to-layout error cannot repair a substantial graph-to-surface error. Even when the full-geodesic classical initializer is held fixed across k, final path-to-surface error changes with the refinement graph in all 84 corresponding $n = 240$ cases. Across the seven tested k values, the within-case range of that error has median 23.05 percentage points, with ranges from 8.05 to 58.31 percentage points across cases.

A single nested sample-size check distinguishes fixed k from fixed k/n. Under geodesic construction and fixed-scale refinement on the base-disk paraboloid, path-to-surface errors are 7.77% for $(n, k) = (240, 8)$, 9.23% for $(480, 8)$, and 2.34% for $(480, 16)$.

Table 4: Small global coordinate residuals can conceal horizontal distortion. Medians over the same three base-disk samples and fits as Table 3. Horizontal and vertical errors use components of one globally aligned residual, normalized by their respective centered truth norms. No separate alignment is performed for either component. Component errors can exceed 100% when the residual exceeds the corresponding truth norm.

Surface	Graph	Global (%)	Horizontal (%)	Vertical (%)
Paraboloid	ambient	0.88	23.66	0.046
Paraboloid	geodesic	1.46	38.58	0.197
Saddle	ambient	0.90	35.24	0.025
Saddle	geodesic	80.38	1751.52	65.930

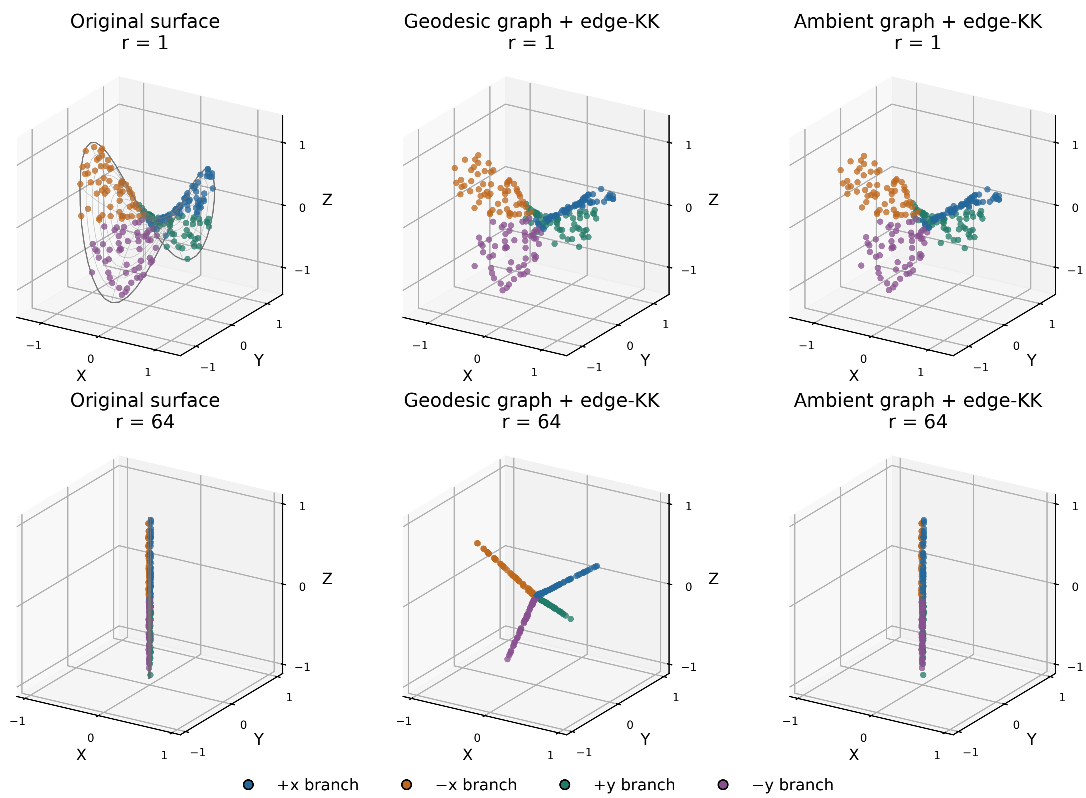

Figure 4: Geodesic and ambient graphs produce different saddle configurations as the domain expands. Shown is the first base-disk sample, $n = 240$, $k = 32$. The fitted columns use stress MDS followed by fixed-scale density continuation. Colors identify the four original height-level branches. The original-surface wireframe includes the full disk boundary. Coordinates are centered, divided by r^2 and aligned; fitted coordinates are additionally divided by their edge calibration factor. Each row uses common limits and equal axis scales (Appendix A.4). At $r = 64$ the original disk's horizontal radius in these units is $1/64$.

Tiny perturbations around stress-MDS starts change the final s_3/s_2 ratios by at most 0.00031 across the tested controls; the corresponding maximum around classical starts is 0.01888. Independently drawn starts can give much larger changes, up to approximately 0.82 under fixed scale. Local stability around an initializer therefore does not imply insensitivity to its selection. Of the 3,648 primary radius-study MDS starts, 25 reach their iteration caps.

5.4 Contraction and fixed-scale controls

Allowing the target scale to vary can produce substantial contraction of the drawing. Across all primary refinements, including both sample sizes, all initializers and both stiffness schedules, the profiled-scale runs attain edge calibration factors as small as 4.07×10^{-10}. Among 6,080 profiled refinements, 830 have calibration below 0.5, 475 below 0.1, and 262 below 0.001; the median is 1.028. The fixed-scale controls range from 0.99339 to 1.02624. Contraction occurs at intermediate radii as well as large ones, so it is not an asymptotic property of the generating surface (Figure 6).

Across the 1,176 primary $n = 240$ graphs, profiled-scale refinement improves path error in 1,016 stress-initialized comparisons, worsens it in 76, and leaves 84 ties. Fixed-scale refinement improves 1,075, worsens 17, and leaves 84 ties. Ties mean absolute changes no larger than 10^{-8} in relative-error units. Neither policy improves path error in every case. Fixed scale removes the contraction mechanism of Proposition 2; nonconvexity and geometric ambiguity remain.

5.5 Density continuation and uniform stiffness

Density continuation is the primary protocol, but the paired uniform-only controls do not show a consistent advantage for it. Holding the initializer, graph, scale policy and total step budget fixed, continuation yields lower retained-path error in 389 of 1,176 fixed-scale stress comparisons, higher error in 541, and ties in 246 (Table 5). The central half of paired changes is small, while the extremes are larger, particularly for coordinate error. Profiled-scale comparisons also favor neither schedule uniformly. The comparisons establish sensitivity to the schedule, not an ordering of the two schedules.

Table 5: Density continuation minus uniform-only error for paired stress-initialized radius-study fits, with 1,176 primary $n = 240$ graphs per scale policy. Negative changes favor continuation. Counts use an absolute tie tolerance of 10^{-8} in relative-error units; intervals and medians are in percentage points (pp). The pairs share graphs and samples and are not independent trials.

Scale	Error	Lower/higher/ tied	Median [quartiles] (pp)	Minimum, maximum (pp)
Profiled	Retained path	411/627/138	0.0000 [-0.0007, 0.0061]	-0.190, 0.349
Profiled	Coordinate	432/614/130	0.0001 [-0.0030, 0.0271]	-1.832, 9.940
Fixed	Retained path	389/541/246	0.0000 [-0.0003, 0.0072]	-0.357, 0.470
Fixed	Coordinate	397/567/212	0.0000 [-0.0044, 0.0412]	-1.952, 9.380

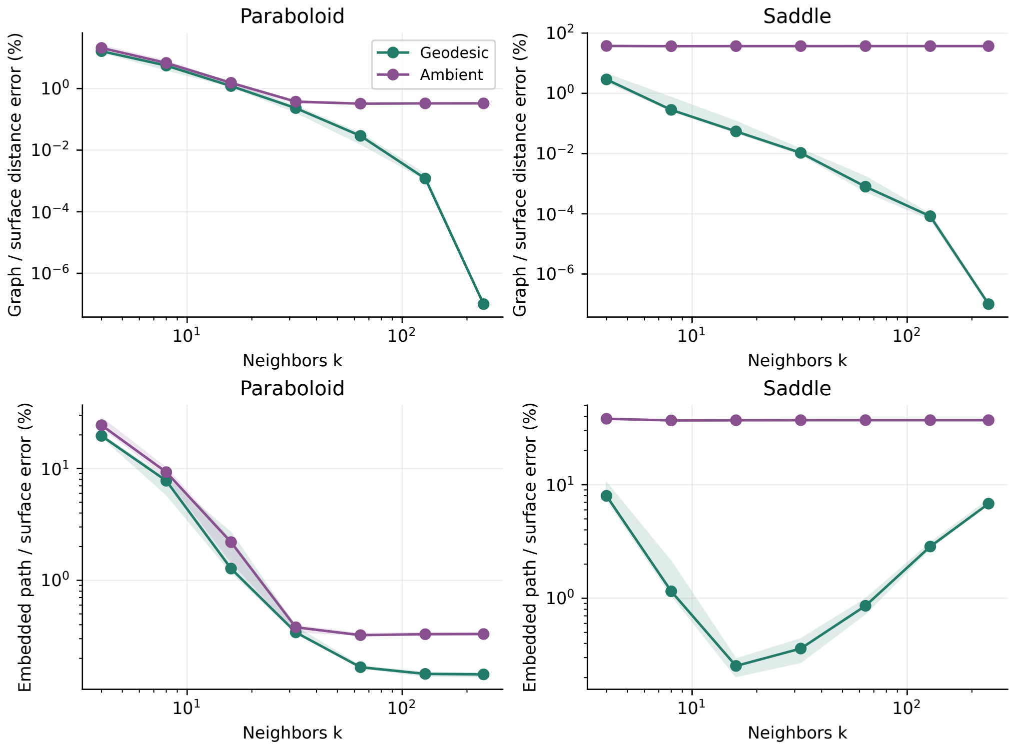

Figure 5: Neighborhood size affects graph-to-surface and embedded-path-to-surface agreement at $r = 64$ under base-disk sampling. Curves are medians and bands observed ranges over three samples. The upper row compares strict graph distances with numerical smooth geodesics. The lower row compares edge-calibrated retained paths after fixed-scale, stress-initialized edge-KK with the same reference. Panels use separate logarithmic ranges; values below $10^{-7}\%$ are drawn at that display floor.

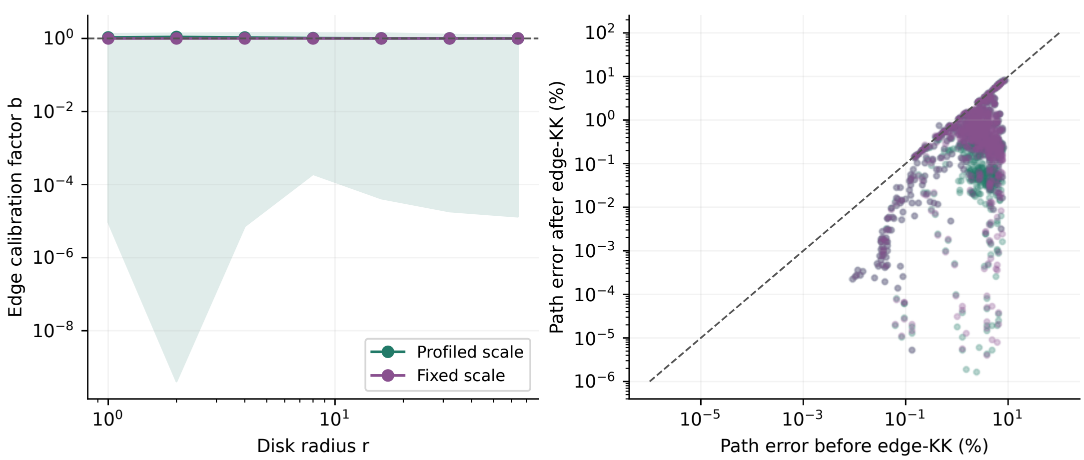

Figure 6: Scale and normalized path fidelity are separate diagnostics. Left: median and full observed range of the edge calibration factor for stress-initialized density continuation across all primary $n = 240$ settings at each radius. Right: paired path errors before and after stress-initialized edge-KK; points below the diagonal improve. Pairs with either error at most 10^{-8} are omitted from this logarithmic scatter only; all comparisons enter the reported counts.

6 Discussion

An embedding is useful only in relation to the measurements it is intended to support, and the experiments show that the usual measurements do not in general certify one another. A small retained-path error does not certify coordinate recovery: in the geodesic saddle fits, accurate lengths along retained graph paths coexist with substantial coordinate error, and path fidelity alone need not fix a configuration. On the flexible three-vertex path of Figure 1, edge lengths are preserved while the angle changes, although this counterexample does not diagnose the dense experimental graphs.

In the ambient regime the original coordinates realize every target edge exactly, and for the three highlighted ambient saddle frameworks at $n = 240$, $r = 64$, $k = 32$, the edge-length Jacobian at the original coordinates has numerical rank $714 = 3n - 6$, with smallest nongauge singular values from 0.00490 to 0.00613 against six rigid-motion values of order 10^{-15}. This supports local infinitesimal rigidity at those frameworks, not global rigidity or a quantitative stability theorem (Gortler et al., 2010), and the experiments do not identify which combination of finite-budget optimization, conditioning and alternative global realizations accounts for each coordinate residual.

Initialization or regularization can select among configurations with similar path errors, so their influence is part of the interpretation.

A small global coordinate residual does not certify intrinsic-distance agreement or transverse geometry either. In the ambient saddle fits, close coordinate agreement coexists with poor path-to-surface agreement, and the same fits have horizontal errors far larger than their global residuals because height dominates the coordinate norm.

Graph-to-layout fidelity must also be distinguished from the graph's approximation to the underlying geometry. When the reference surface is unknown, agreement with the graph can assess the embedding step but cannot verify the scientific meaning of the graph.

The surface limits explain one source of these separations. At the height scale, the paraboloid loses angular separation and approaches a line metric. The saddle retains the separation of four height-level branches and approaches a tree metric. Under the stated rank condition, classical MDS represents the latter with three nonzero limiting directions. A one-dimensional intrinsic structure therefore need not give a one-dimensional Euclidean embedding. The raw-stress fits also retain broad three-dimensional span in the tested saddle settings, although a universal raw-stress result has not been established. Curvature decay or a quadratic fit to the displayed coordinates alone cannot distinguish these distance geometries.

A decreasing objective must likewise be read with its scale convention. Fixed target scale removes the contraction mechanism of Proposition 2, but the observed path-error increases show that it neither ensures improvement in every graph nor eliminates dependence on initialization.

For biological observations, these distinctions affect how a branch, gap or sequence of nearby samples can be interpreted. Stability to plausible graph and embedding choices can support an interpretation of the drawing, while biological evidence must establish what those relationships mean. The synthetic examples isolate geometric effects; they do not establish biological trajectories or validate features of a particular microbiome dataset.

The comparisons concern two quadratic surface families and finite optimization budgets. In the primary radius study, domain size increases with sample size held fixed, so the samples do not become increasingly dense surface meshes. Surface-area and base-disk sampling also assign different weights to the geometry. The bounded-saddle study differs from the radius study in domain, sample size and scale policy, and their scores cannot be pooled. Within these designs, the separate path, endpoint, scale and coordinate diagnostics identify which aspects of an embedding improve and which remain undetermined.

A Computational details

A.1 Sampling and geodesic references

Uniform sampling in the base disk and on the surface weights radial positions differently. For both radius-study surfaces the area element in polar base coordinates is $\rho\sqrt{1+4\rho^2}\,d\rho\,d\theta$. Thus surface-area sampling uses the radial cumulative distribution

$$\Psi_r(\rho) = \frac{(1+4\rho^2)^{3/2}-1}{(1+4r^2)^{3/2}-1}, \qquad 0 \le \rho \le r,$$

with independent uniform angle; base-disk sampling uses $\Psi_r(\rho) = \rho^2/r^2$. The same uniform draws are transformed at each radius. The primary NumPy generator seeds are $20260904+j$ for replicate j. The nested additional 240 observations use seed 20261905. Corresponding surface/measure settings share these draws, so settings are paired rather than independent samples.

Paraboloid distances use the radial first integral of the geodesic equations, with separate turning and monotone radial paths and 62 bisection steps for the angular boundary condition. Saddle distances use shooting for the Cartesian geodesic equations in base coordinates scaled by r, with continuation from smaller radii. The primary endpoint tolerance is 2×10^{-9} in these base units; the integration relative tolerance is 2×10^{-10}. High-radius validation uses tighter shooting and independent collocation/integration on the saddle, and refined radial solves with adaptive quadrature on the paraboloid. The bounded-saddle calibration uses a separate triangulated-surface reference, with mesh-refinement and smooth-geodesic boundary-value checks recorded in its original protocol.

A.2 Why complete-surface geodesics stay inside the disk

The smooth solvers do not impose a reflecting or obstacle boundary at the disk edge. This agrees with the stated surface-patch distance. To see this, write $f(x,y) = x^2 + \eta y^2$, $\eta \in \{-1, 1\}$, and let $u = (x, y)$ be the base projection of an affinely parametrized surface geodesic, with $v = u'$ and $\rho^2 = x^2 + y^2$. The graph geodesic equation gives

$$u'' = -\frac{4(v_x^2 + \eta v_y^2)}{1+4\rho^2}(x, \eta y),$$

so

$$(\rho^2)'' = 2\|v\|^2 - \frac{8(x^2+\eta y^2)(v_x^2+\eta v_y^2)}{1+4\rho^2} \ge \frac{2\|v\|^2}{1+4\rho^2} \ge 0.$$

Thus the squared radius is convex along a geodesic segment and never exceeds the larger endpoint squared radius. A minimizing geodesic on the complete surface therefore stays in the disk patch whenever its endpoints do.

Both full graphs are complete: their induced metric dominates the Euclidean base metric, and on a bounded base region the two metrics are locally uniformly comparable. The saddle is also simply connected and has negative Gaussian curvature. The Hadamard–Cartan theorem therefore makes its geodesic between two points unique and minimizing (Gorodski, 2022, Theorem 6.5.2, Corollary 6.5.3, and the proof of Lemma 6.5.4, pp. 124–125). This justifies the saddle shooting target; it does not make numerical shooting error zero. The positive curvature of the paraboloid does not permit the same uniqueness argument.

A.3 Stiffness and optimization

The density continuation initially gives more weight to edge lengths that are common in the input graph, then moves to equal edge weights. For each graph, the density of positive input edge lengths is estimated by a Gaussian kernel, denoted $\widehat{f}$, computed using R's `density` with `nrd0` bandwidth and 512 grid points from the minimum to maximum length. Linear interpolation evaluates the density at each edge length. Dividing by the mean of these values gives the initial relative weights $w_e = \widehat{f}(\ell_e)/(|E|^{-1} \sum_f \widehat{f}(\ell_f))$. The stiffness at continuation value α is

$$\kappa_e(\alpha) = (1-\alpha)w_e + \alpha, \qquad \alpha \in \{0, 1/4, 1/2, 3/4, 1\}.$$

The implementation normalizes stiffnesses to mean one; no additional signal transformation or positive clipping floor is applied. Equal lengths fall back to uniform stiffness. Uniform-only controls use $\alpha = 1$ for the full budget.

Edge-KK uses gradient descent with Armijo backtracking: initial step one, step multiplier 1/2, Armijo coefficient 10^{-4}, minimum step 10^{-8}, and Euclidean gradient-norm stopping threshold 10^{-8}. Accepted coordinates are recentered. The edge-length smoothing parameter is zero in both studies. These absolute stopping rules act in each study's fitting units, which is relevant to the contraction analysis. Stress MDS uses SMACOF tolerance 10^{-8} in both primary studies. Radius-study MDS random starts use seed $8300000 + 1000j$, and their draw is reused across graph settings within replicate j.

Neighbor ranks in the radius study order by distance and then vertex index; an edge is present if either endpoint selects the other. Graph edges are canonically ordered before path preparation. The single-route cache records one route per pair. Floating-point near ties in path preparation can yield route lengths slightly different from the strict shortest-distance matrix; the experiments therefore score retained paths against their recorded input route lengths and chords against the strict matrix. Source manifests and the original validators retain this distinction. The mathematical statements for exact shortest paths do not require these discrepancies to be treated as zero.

A.4 Rendering of Figure 4

All coordinates are centered and divided uniformly by r^2; fitted coordinates are also divided by their edge calibration factor and orthogonally aligned to the original observations. Each row uses common limits, equal spatial axes and an orthographic view. No coordinate direction is enlarged independently, so the narrow appearance of the original disk at $r = 64$ reflects its horizontal radius of 1/64 in these units.

Data and code availability

The manuscript's figures and tables are generated from versioned exports of the two experiments, both run with the `grip` R package. A source manifest records the exact commit and SHA-256 hashes of source and exported files. Portable verification and rendering scripts accompany the manuscript. Full fitting protocols, input matrices, coordinates and validators remain in the experiment directories specified by that manifest in the grip repository. The bounded study used grip 0.2.0.9000 and the radius study 0.2.0.9001; both used smacof 2.1.7. The manifest retains their distinct provenance.

Acknowledgments and author review

The manuscript was prepared with AI assistance for code, derivation checks, evidence organization and drafting. The authors retain responsibility for the scientific content. This author-review version has not been submitted; final authorship, disclosure wording and scientific approval remain with the authors.

References

Reyan Ahmed, Cesim Erten, Stephen Kobourov, Jonah Lotz, Jacob Miller, and Hamlet Taraz. Size should not matter: Evaluating network visualizations with stress. arXiv:2408.04688v2, 2026. URL https://arxiv.org/html/2408.04688v2. Version 2, July 22, 2026; first version 2024.

Claudio Gorodski. An introduction to Riemannian geometry, 2022. URL https://www.ime.usp.br/~gorodski/teaching/mat5771-2024/master02-21-2024.pdf. Preliminary version 3, June 27, 2022; author's notes hosted at Universidade de São Paulo.

Steven J. Gortler, Alexander D. Healy, and Dylan P. Thurston. Characterizing generic global rigidity. *American Journal of Mathematics*, 132(4):897–939, 2010. doi: 10.1353/ajm.0.0132. URL https://arxiv.org/pdf/0710.0926v5. Source verification uses author preprint version 5.

Patric Hagmann, Leila Cammoun, Xavier Gigandet, Reto Meuli, Christopher J. Honey, Van J. Wedeen, and Olaf Sporns. Mapping the structural core of human cerebral cortex. *PLOS Biology*, 6(7):e159, 2008. doi: 10.1371/journal.pbio.0060159. URL https://doi.org/10.1371/journal.pbio.0060159.

Tomihisa Kamada and Satoru Kawai. An algorithm for drawing general undirected graphs. *Information Processing Letters*, 31(1):7–15, 1989. doi: 10.1016/0020-0190(89)90102-6.

Sunhyuk Lim and Facundo Mémoli. Classical multidimensional scaling on metric measure spaces. *Information and Inference: A Journal of the IMA*, 13(2):iaae007, 2024. doi: 10.1093/imaiai/iaae007. URL https://academic.oup.com/imaiai/article/13/2/iaae007/7667009. Source verification uses author preprint arXiv:2201.09385v6, May 13, 2024.

Patrick Mair, Patrick J. F. Groenen, and Jan de Leeuw. More on multidimensional scaling and unfolding in R: smacof version 2. *Journal of Statistical Software*, 102(10):1–47, 2022. doi: 10.18637/jss.v102.i10. URL https://www.jstatsoft.org/article/view/v102i10.

Facundo Mémoli, Osman Berat Okutan, and Qingsong Wang. Metric graph approximations of geodesic spaces. arXiv:1809.05566v5, 2023. URL https://arxiv.org/pdf/1809.05566v5. Version 5, October 26, 2023; first version 2018.

M. E. J. Newman. The structure of scientific collaboration networks. *Proceedings of the National Academy of Sciences*, 98(2):404–409, 2001. doi: 10.1073/pnas.98.2.404. URL https://doi.org/10.1073/pnas.98.2.404.

R Core Team. *Classical (Metric) Multidimensional Scaling: cmdscale*. R Project for Statistical Computing, 2026. URL https://stat.ethz.ch/R-manual/R-devel/library/stats/html/cmdscale.html. R-devel stats documentation, version 4.6.0; accessed 2026-09-05.

Joshua B. Tenenbaum, Vin de Silva, and John C. Langford. A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323, 2000. doi: 10.1126/science.290.5500.2319. URL https://graphics.stanford.edu/courses/cs233-20-spring/ReferencedPapers/isomap.pdf.

F. Alexander Wolf, Fiona K. Hamey, Mireya Plass, Jordi Solana, Joakim S. Dahlin, Berthold Göttgens, Nikolaus Rajewsky, Lukas Simon, and Fabian J. Theis. PAGA: graph abstraction reconciles clustering with trajectory inference through a topology preserving map of single cells. *Genome Biology*, 20:59, 2019. doi: 10.1186/s13059-019-1663-x. URL https://doi.org/10.1186/s13059-019-1663-x.